

Arrays

1. Subarray sum (Hashing)
2. Sort 0,1,2 (DNF)
3. Majority Element (Moore's Voting algorithm)

***** How to think where this question should go *****

Suppose you found a brute force solution to a problem, and its taking linear time ($O(n)$) and giving TLE. And now to reduce TC you need to make it $\log(n)$ or constant.

As soon as you thought of $\log(n)$, two cases arises binary search and priority queue.

***** Merge Sort *****

Jb bhi tumhe lge ki agar ye array sorted hota to mai easily solve kr leta, lekin tum uss array ko sort nhi kr skte kyunki tumhe order maintain krna hai aise cases me try using merge sort.

***** Binary Search *****

Agar array sorted hai to binary search. Pr tum left jaoge ya right uski condition change hoti rahegi according to question(Bs yahi condition pata lagani hai).

For example lower_bound me alag hogi, upper_bound me alag hogi.

Ek aur technique, kai bar while solving tum ek ans variable loge aur koi particular condition met hogi to kahoge maybe this is the possible answer aur use store kar

loge aur fir agle loop me wo condition dobara met hoti hai iska mtlb it is a better answer aur ab tum ise store kr loge aur age badhoge.

***** Rotated Binary Search *****

Condition: Ek side humesha sorted hogi, uske basis pr hum check karenge kis side pr move krna hai.

***** Binary Search on Answers *****

You will always have a range of possible answers.

For example in square root of n , if n is 28. Your possible answer can be 1,2,3,4,5.

This question in other words. Maximum integer which on squaring $\leq n$.

$1*1 \leq 28$, $2*2 \leq 28$, ... $5*5 \leq 28$. (Possible answers, since we need max, the answer is 5)

$6*6 \leq 28$, ... (Not possible answers).

In Binary Search on Answers we will always have possible and not possible answers and we have to choose a max or min from that possible range.

In questions ke liye, sbse Pehle tum brute force sochoge ki 1 se start krta

hu(or according to question) aur n tak traverse krke check kr leta hu.
Sbse pehli possible value min de degi ya fir upar wale example me aakhri value jo answer degi wo tumhara answer hoga.

Ab yha mai low = 1(ya according to question lunga) and high = max or sum of array lunga aur Binary search lga dunga.

Suppose kisi question ke liye 1-20 le beech me answer lie karega. 1-4(Not possible wali range hai), (5-20) possible wali range hai.
low = 1 hojayega aur high = 20 ho jayega.
mid = 10 aayega, tum kahoge okay this is can be the possible answer but aur better answer left me exist kr skta hai.

***** Binary Search on Min(Max) or Max(Min) *****

LEETCODE 410: SPLIT ARRAY LARGEST SUM

Ye question BS ke 3rd type ka hai. Binary Search on answer jis me maximum sum ko minimize krna hai.

Jaise hi hume dikha ki maximum sum ko minimize krna hai hume pata chal gaya BS on answer lagega.

Pr humne sochna Brute Force se start kiya. Humne socha maximum sum(jo humara answer hoga wo kaha se kaha tak lie krega).
Hume pata chala 1 se lekar total sum tak lie kr skta hai jb k == 1 hoga aur hume poora subarray lena padega.

Ab hum brute force apnayenge aur 1 se start karenge aur total sum tak jayenge. Aur har value ke liye check karenge ki is maximum sum ke liye kitne subarrays banaye ja skte hai.

Iss example ke liye 7,2,5,10,8.

Suppose humne 1 se start kara aur hum 9 par pahuche. Ab 9 ke liye check karenge ki kitne subarrays possible hai.

1st subarray: 7, 2

2nd subarray: 5

3rd subarray: 10(Not possible kyunki ye to khud hi maximum possible sum se bada hai)

Aise hi hum 18 par pahuchenge

1st subarray: 7,2,5

2nd subarray: 10,8

Yha pr hume pata chala ki 18 max sum ke liye hum 2 subarrays bana skte hai, jo ki k ke baraabar hai, hum true return karenge aur ye humara answer hoga.

Yha tk to humari Brute Force thi, Ab Binary Search.

low = 1, high = sum, ans = -1

For mid:

1. Agar count <= k aa raha hai iska matlab subarray ka sum itna hai ki humare paas 'k' subarrays easily ban skte hai aur ye humara possible answer ho skta hai

aur hume minimum chahiye is liye left move karenge(high = mid - 1)

Why count <= k not count == k?

Agar ye example hota 2,3,1,1,1,1,1. Tb max sum 3 ke liye aur k = 5

ke liye

1st subarray: 2

2nd subarray: 3
3rd subarray: 1 1 1
4th subarray: 1 1

Bhale hi ye 4 subarray me aa gaya pr hum chahte to

1st subarray: 2
2nd subarray: 3
3rd subarray: 1 1 1
4th subarray: 1
5th subarray: 1

2. Agar $\text{count} > k$ aa rha hai iska matlab $\text{mid}(\text{subarray ka sum})$ itna kam hai ki uss particular subarray ke liye bohot kam elements include ho pa rahe hai jiski wajah se subarray ka size 'k' se exceed ho ja rha hai. Isliye subarray ka sum badhana hai, hence, hum right me move kr jayenge($\text{low} = \text{mid} + 1$)